

Project Artefacts Requirements

Edmonton Residential Real Estate — Data Science Lifecycle Documentation

Group 5 | DATA 3960 | NAIT | April 2026

1. Data Sampling and Collection

» *How were the data collected?*

The primary dataset is the Edmonton residential sold transactions file (2000–2025), provided as a compressed ZIP archive and extracted using Python's zipfile library. Loaded into pandas with **latin1 encoding** (multi-system municipal data entry) and **low_memory=False** (full dtype inference on 515,122 records). Three supplementary datasets were integrated: Bank of Canada Series V80691335 (1,362 weekly mortgage rate observations), Alberta CPI from Statistics Canada Table 18-10-0004-01 (annual, base 2024=124.5), and the City of Edmonton Property Assessment file for geocoding and assessed value enrichment.

» *Was there any potential bias introduced in the sampling process?*

- **Transaction selection bias:** Only completed arm's-length market transactions are captured. Off-market sales, foreclosures, and intra-family transfers are absent.
- **LINC identifier drift bias:** Administrative changes cause properties with long histories to appear artificially new, biasing appreciation toward shorter price series. Corrected by the Base_Address reconciliation pipeline (234,486 drift instances resolved).
- **Geocoding confidence bias:** 83,148 records (16.1%) received IMPUTED geocoding. Geo_Confidence metadata enables downstream filtering by spatial precision tier.

2. Data Cleaning

» *What type of data and what is the granularity?*

Mixed-type tabular data: numeric (prices, rates, area), categorical (neighbourhood, asset type), boolean (Is_Repeat_Transaction), datetime (Sold_Date), and string (address, LINC). Granularity is **transaction-level** — one row per completed residential sale, daily temporal resolution, 2000–2025.

» *Distribution, outliers, missing or invalid entries?*

Sold_Price is strongly right-skewed (median \$305,000, max \$18.4M). Log transformation applied before OLS regression. Key cleaning actions:

- 18 records with corrupt dates (e.g. year 2992) removed via errors='coerce'
- 9,246 ghost assets (Sold_Price > \$100K, FlrArea ≤ 1 sqft) reclassified as Pre-Construction/Land
- 22,311 near-duplicate repeat transactions removed (< 90 days, < 2% price change)
- 1,618 appreciation outliers capped at ±500%; 29 early Mortgage_Rate nulls filled with 8.25%
- 9,357 records reclassified via Oliver → Wihkwêntôwin neighbourhood crosswalk

3. Exploratory Data Analysis

» *Is there any correlation between variables?*

Three primary relationships with Sold_Price were identified: (1) **Floor area** — strong positive, non-linear (Log_FlrArea used in OLS); (2) **Sale_Year** — strong positive due to 25-year upward trend (VIF=560.51 — collinearity flagged as limitation); (3) **Mortgage_Rate** — Pearson R = -0.238 in the Macro segment (p < 0.001), confirming interest rates act as a price ceiling. Rolling_3Yr_Neighbourhood_Growth ranged -53.8% to +233.0%, confirming substantial local heterogeneity beyond the macro signal.

» *How would you visualize relationships among variables?*

Eight PNG outputs were produced from the modeling notebook. Key visualizations: **Group5_RQ2_Drivers.png** — dual-axis mortgage rate vs median price (25-year line+bar); **Group5_OLS_Diagnostics.png** — residuals vs fitted, Q-Q plot, residual std by price band; **Group5_ActualVsPredicted.png** — scatter plots for all three models with diagonal reference line; **Group5_RQ3_SARIMA.png** — Model A vs Model B forecasts with 95% CI on 2024–2025 holdout. Interactive versions accessible via the Streamlit dashboard (app.py) and Power BI report with slicers for year, neighbourhood, and rate environment.

4. Data Modeling and Inferences

» *What type of ML problem? What model and why?*

This is a **supervised regression problem** — target variable `Sold_Price` is continuous. Three models were trained comparatively:

- **OLS Linear Regression (baseline):** Interpretable; coefficient significance testable via t-statistics. `Mortgage_Rate` coef = -0.116 ($p < 0.001$). $R^2 = 0.952$.
- **Random Forest (200 trees, max_depth=15):** Captures non-linear interactions without explicit feature engineering. Reduces variance by averaging independent trees. $R^2 = 0.983$.
- **Gradient Boosting (200 estimators, lr=0.05, subsample=0.8) — Best:** Reduces bias via sequential residual correction (stagewise fitting). Shrinkage ($lr=0.05$) prevents outlier overfitting. Outperforms RF on structured tabular data — bias reduction more valuable than variance reduction when signal is systematic. $R^2 = 0.9922$, MAE = \$16,536.

» *How do you engineer features? What are the rationales?*

Sixteen features across four categories. The five pre-modeling engineered features (F1–F5) directly connect pipeline outputs to model inputs:

- **Macroeconomic (`Mortgage_Rate`, `Phase_Encoded`, `Context_Encoded`, `Driver_Encoded`):** Encodes interest rate ceiling effect and 8 named BoC rate regimes so the model distinguishes 2022 Rate Shock from 2010 Low Rate era
- **Temporal (`Sale_Year`, `Sale_Month`, `Is_Spring_Market`):** Captures 25-year trend and Edmonton's spring market peak (March–June)
- **Property (`Log_FlrArea`, `Bedrms_AG`, `Full_Baths`, `DOM`, `Property_Age`):** Log transform handles non-linear price-size relationship; DOM is a market liquidity signal
- **Engineered F1–F5 (`Rolling_3Yr_Growth`, `Real_Price_Index`, `Days_Since_Last_Sale`, `Appreciation_%`):** Embeds inflation context, neighbourhood momentum, and repeat-sale appreciation directly into the model feature space

» *How do you perform cross-validation?*

Standard k-fold cross-validation was deliberately not applied — shuffling allows future months to appear in training folds, introducing **temporal leakage**. A **temporal holdout split** was used: training on 418,302 records (2000–2023), evaluating on 54,818 records (2024–2025). This is the methodologically correct strategy for time-series regression, simulating realistic deployment where only past data is available at prediction time.

» *Bias-variance tradeoff, overfitting, and model improvement?*

VIF analysis confirmed extreme multicollinearity (`Sale_Year` VIF=560.51). Training $R^2 = 0.9998$ vs test $R^2 = 0.9922$ indicates **mild overfitting**. Residual heteroskedasticity confirmed — MAE grows from \$7,990 (<\$300K) to \$59,865 (>\$1M), visualized in `Group5_ResidualByBand.png`. OLS diagnostics: Shapiro-Wilk $W=0.931$ ($p < 0.001$, non-normal residuals), Breusch-Pagan $p < 0.001$ (heteroskedastic). Cook's Distance: 13,733 high-influence points (4.9%). **Improvements:** remove `Sale_Year` to eliminate VIF=560 collinearity; apply systematic hyperparameter tuning (GridSearchCV); add spatial proximity features using HIGH/MEDIUM geocoded records; single-family filter for repeat-sale features; retrain annually.

4.1 Inferential Analysis — Hypothesis Tests ($\alpha = 0.05$)

RQ1 — Levene's Test: H_0 : $\text{var}(\text{reconciled RSI}) = \text{var}(\text{broken RSI})$. $F = 1.33$, $p = 0.25$ — fail to reject H_0 at 5%. Directional finding confirmed by bootstrap 95% CIs: reconciled [11.87%, 21.47%] vs broken [8.96%, 17.16%]. Higher reconciled volatility is economically meaningful — captures genuine cross-cycle appreciation swings invisible to the broken dataset.

RQ2 — One-Way ANOVA + Tukey HSD: H_0 : prices do not differ across market cycle phases. $F = 16,194.68$ ($p < 0.001$) — reject H_0 . Effect size: Cohen's $f = 0.484$ (large). All 28 Tukey HSD pairwise comparisons significant. Pearson $R = -0.238$ ($p < 0.001$) in Macro segment tests interest rate ceiling hypothesis. OLS `Mortgage_Rate` coefficient = -0.116 ($p < 0.001$) quantifies causal magnitude controlling for property attributes. Endogeneity acknowledged: BoC rates treated as exogenous but may partially respond to housing market conditions.

RQ3 — ADF + SARIMA: ADF at level: $p = 0.164$ (non-stationary). At first difference: $p = 0.003$ — reject H_0 , confirming $d=1$. AIC grid search over SARIMA($p,1,q$)(1,1,1,12) for $p,q \in \{0,1,2\} \rightarrow (2,1,2)$ optimal (AIC=5,452.71). Ljung-Box $p = 0.88$ — no residual autocorrelation. Model B underperforms Seasonal Naïve (MAE \$47,260 vs \$23,652) in the stable 2024–2025 period — honest finding: SARIMA adds value in volatile periods. Both models outperform Random Walk (+5.6% and +25.1% respectively), confirming genuine forecasting value.